MS&E 342: Stochastic Systems and Learning Theory with Applications in Finance

Lecture 2-3: Stochastic Foundations of Generative Diffusion Models

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This note is based on *Score-Based Diffusion Models via Stochastic Differential Equations* [6], which provides a SDE/probabilistic view of generative diffusion models.

The main message is simple:

- the *forward* diffusion adds noise to data, which is of our design;
- the *time reverse* diffusion has a drift determined by the score $\nabla \log p_t$;
- the score can be learned from noisy data by denoising score matching;
- once the score is learned, we can sample new data either by a time-reverse SDE or by the probability flow ODE.

Throughout, we assume enough regularity so that all densities exist, are smooth and strictly positive, and all integrations by parts and uniqueness statements for the associated PDEs are justified. The purpose here is clarity of structure rather than maximal generality.

Contents

1 Motivation and setup

1.1 Motivation

In recent years, diffusion-based generative models have emerged as one of the most successful paradigms for high-dimensional data generation. Starting from the seminal work of [4, 5], these models have achieved state-of-the-art performance across a wide range of domains, including image synthesis, audio generation, molecular design, and scientific simulation.

A key reason for this success is that diffusion models convert a difficult *global density estimation problem* into a sequence of *local denoising problems*. Instead of directly parameterizing the data distribution p_{data}, which is typically highly non-Gaussian and concentrated on a low-dimensional manifold, diffusion models construct a path of intermediate distributions $(p_t)_{t\in[0,T]}$ that gradually interpolate between p_{data} and a simple reference law (typically Gaussian). This path is governed by a stochastic differential equation, which provides both analytical tractability and algorithmic flexibility.

From a modeling perspective, diffusion models enjoy several structural advantages:

- **Likelihood-free yet principled:** The model is trained via score matching, avoiding explicit likelihood evaluation while remaining grounded in well-defined variational objectives.
- **Stability in high dimension:** Unlike GANs, diffusion models do not rely on adversarial training and are therefore significantly more stable in practice.
- **White-box structure:** The dynamics are explicitly described by SDEs and PDEs (Fokker–Planck), making it possible to incorporate domain knowledge (e.g., constraints, invariances, or financial structure) directly into the model.
- **Flexible sampling:** The same learned object (the score) can be used to generate samples via either stochastic dynamics (reverse SDE) or deterministic flows (probability flow ODE).

From a mathematical viewpoint, diffusion models are particularly appealing because they admit a clean formulation in terms of stochastic processes and partial differential equations. The entire framework can be understood through three fundamental components:

1. a forward diffusion that transforms data into noise;
2. a time-reversal identity that expresses the backward dynamics in terms of the score $\nabla \log p_t$;
3. a statistical procedure (denoising score matching) that allows us to estimate this score from noisy observations.

This structure not only explains the empirical success of diffusion models, but also makes them a natural candidate for applications where interpretability, stability, and incorporation of constraints are essential — such as in finance, where data are scarce, heavy-tailed, and subject to structural constraints.

1.2 Representative applications of diffusion models

We highlight several representative applications that illustrate both the empirical success and the modeling flexibility of diffusion-based methods.

(1) Image generation and editing. Diffusion models have achieved state-of-the-art performance in image synthesis, as demonstrated by models such as DDPM [8] and Stable Diffusion [9]. These models generate high-resolution, photorealistic images and support conditional generation (e.g., text-to-image).

A notable feature is their ability to perform *controlled generation*, such as inpainting, super-resolution, and style transfer, by modifying the reverse dynamics.

(2) Scientific machine learning and physical systems. Diffusion models have been successfully applied to scientific domains, including molecular generation and protein folding. For example, [10] uses diffusion models to generate molecular structures with desired properties.

The continuous-time formulation makes it natural to incorporate physical constraints and symmetries, which is critical in scientific applications.

(3) Time series and stochastic dynamics. Recent works extend diffusion models to sequential and time-series data, including financial time series. These models can capture complex temporal dependencies and heavy-tailed behavior, making them suitable for applications such as scenario generation and stress testing.

(4) Conditional generation and inverse problems. Diffusion models are particularly effective for solving inverse problems by conditioning the reverse dynamics. Examples include image reconstruction, data imputation, and posterior sampling in Bayesian inference.

These applications share a common feature: diffusion models provide a *probabilistic dynamical system* perspective on data generation, where learning reduces to estimating the score field that governs the evolution of distributions.

1.3 Mathematical set-up

The goal of generative modeling is to sample from a complicated data distribution p_{data} on $\mathbb{R}^d$. Directly modeling p_{data} is hard when the data live in high dimension and are concentrated on a highly non-Gaussian set. Diffusion models avoid this difficulty by introducing a whole family of intermediate laws $(p_t)_{0\le t\le T}$.

The strategy is to solve the easy problem first:

1. Start from data $X_0 \sim p_{\text{data}}$.
2. Add noise gradually, so that X_T is close to a simple reference law p_{noise}.
3. Learn how to reverse this noising procedure.

Why is this helpful? Because the forward noising step is chosen by us, so it is analytically simple. In particular, for the standard models, the conditional law of X_t given X_0 is Gaussian. That makes the score-learning problem much more tractable.

We therefore begin with the forward SDE

$$\mathrm{d}X_t = f(t, X_t)\ \mathrm{d}t + g(t)\ \mathrm{d}W_t, \qquad X_0 \sim p_{\text{data}}, \qquad 0 \le t \le T, \tag{1}$$

where:

- $f : [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$ is the drift,
- $g : [0, T] \to (0, \infty)$ is the scalar noise level,
- p_t denotes the density of X_t.

The density-level evolution is the Fokker–Planck equation.

Proposition 1.1 (Fokker–Planck equation)**.** *If X_t solves (1), then its density satisfies*

$$\partial_t p_t(x) = -\nabla\cdot\big(f(t, x)p_t(x)\big) + \frac{1}{2}g(t)^2\Delta p_t(x). \tag{2}$$

Proof. Let $\varphi \in C_c^\infty(\mathbb{R}^d)$ be a test function. By Itô's formula,

$$\varphi(X_t) - \varphi(X_0) = \int_0^t \Big(f(s, X_s) \cdot \nabla\varphi(X_s) + \tfrac{1}{2}g(s)^2\Delta\varphi(X_s)\Big)\ \mathrm{d}s + M_t,$$

where M_t is a martingale. Taking expectations and differentiating in t gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\varphi(X_t)] = \mathbb{E}\Big[f(t, X_t) \cdot \nabla\varphi(X_t) + \tfrac{1}{2}g(t)^2\Delta\varphi(X_t)\Big].$$

Since X_t has density p_t,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} \varphi(x)p_t(x)\ \mathrm{d}x = \int_{\mathbb{R}^d} \Big(f(t, x) \cdot \nabla\varphi(x) + \tfrac{1}{2}g(t)^2\Delta\varphi(x)\Big)p_t(x)\ \mathrm{d}x.$$

Integrating by parts moves the derivatives from φ onto p_t and yields (2). □

Equation (2) is the starting point for everything that follows. The reverse-time formula, the ODE sampler, and the convergence bounds are all consequences of comparing PDEs.

2 Time reversal

We now turn to the key structural fact behind diffusion models: the forward diffusion can be run backward, and the backward drift is determined by the score.

Fix $T > 0$, and define a backward process started from the terminal law:

$$Y_0 \sim p_T.$$

We seek a dynamics for Y_t whose law at time t is exactly p_{T-t}. This is the formulation relevant for sampling: if we can evolve p_T backward to $p_0 = p_{\text{data}}$, then we know what dynamics to imitate when starting from noise.

Theorem 2.1 (Time-reversal formula, marginal version)**.** *Let q_t denote the density of the SDE*

$$\mathrm{d}Y_t = \Big(- f(T-t, Y_t) + g(T-t)^2 \nabla \log p_{T-t}(Y_t) \Big)\, \mathrm{d}t + g(T-t)\; \mathrm{d}\overline{W}_t, \qquad Y_0 \sim p_T, \tag{3}$$

where $\overline{W}$ is a Brownian motion. Then

$$q_t = p_{T-t} \qquad \textit{for every } 0 \le t \le T.$$

In particular, $Y_T \sim p_0 = p_{\mathrm{data}}$.

Proof. Let q_t denote the density of Y_t. Since Y solves

$$dY_t = \Big(-f(T-t, Y_t) + g(T-t)^2 \nabla \log p_{T-t}(Y_t)\Big)\, dt + g(T-t)\, d\overline{W}_t, \qquad Y_0 \sim p_T,$$

the density q_t satisfies the Fokker–Planck equation

$$\partial_t q_t(x) = -\nabla\cdot \Big(\left[-f(T-t, x) + g(T-t)^2 \nabla \log p_{T-t}(x)\right] q_t(x) \Big) + \frac{1}{2} g(T-t)^2 \Delta q_t(x), \tag{4}$$

with initial condition

$$q_0(x) = p_T(x).$$

We now verify directly that the function

$$q_t(x) = p_{T-t}(x)$$

satisfies (4). Differentiating in t and using the forward Fokker–Planck equation for p_s,

$$\partial_s p_s(x) = -\nabla\cdot (f(s, x) p_s(x)) + \frac{1}{2} g(s)^2 \Delta p_s(x),$$

we obtain

$$\begin{aligned}
\partial_t q_t(x) &= -\partial_s p_s(x)\big|_{s=T-t} \\
&= \nabla\cdot (f(T-t, x) p_{T-t}(x)) - \frac{1}{2} g(T-t)^2 \Delta p_{T-t}(x).
\end{aligned}$$

Since $q_t = p_{T-t}$, this becomes

$$\partial_t q_t(x) = \nabla\cdot (f(T-t, x) q_t(x)) - \frac{1}{2} g(T-t)^2 \Delta q_t(x). \tag{5}$$

On the other hand, the right-hand side of (4), evaluated at $q_t = p_{T-t}$, is

$$\begin{aligned}
&- \nabla\cdot \Big(\left[-f(T-t, x) + g(T-t)^2 \nabla \log p_{T-t}(x)\right] q_t(x) \Big) + \frac{1}{2} g(T-t)^2 \Delta q_t(x) \\
&\quad = \nabla\cdot (f(T-t, x) q_t(x)) - \nabla\cdot \left(g(T-t)^2 q_t(x) \nabla \log p_{T-t}(x)\right) + \frac{1}{2} g(T-t)^2 \Delta q_t(x).
\end{aligned}$$

Now $g(T-t)^2$ is independent of x, and since $q_t = p_{T-t}$,

$$q_t(x) \nabla \log p_{T-t}(x) = p_{T-t}(x) \nabla \log p_{T-t}(x) = \nabla p_{T-t}(x) = \nabla q_t(x).$$

Therefore

$$\nabla\cdot\big(g(T-t)^2 q_t(x)\nabla\log p_{T-t}(x)\big) = g(T-t)^2\Delta q_t(x),$$

and the right-hand side becomes

$$\nabla\cdot\big(f(T-t,x)q_t(x)\big) - g(T-t)^2\Delta q_t(x) + \frac{1}{2}g(T-t)^2\Delta q_t(x) = \nabla\cdot\big(f(T-t,x)q_t(x)\big) - \frac{1}{2}g(T-t)^2\Delta q_t(x).$$

This coincides exactly with (5).

Finally, at $t=0$,

$$q_0(x) = p_{T-0}(x) = p_T(x),$$

which matches the initial law of Y_0. Hence $q_t = p_{T-t}$ satisfies the Fokker–Planck equation and the initial condition. By uniqueness of solutions to the Fokker–Planck equation, we conclude that

$$q_t = p_{T-t} \qquad \text{for all } 0 \le t \le T.$$

In particular,

$$Y_T \sim q_T = p_0 = p_{\text{data}}.$$

□

Remark 2.2 (Stein's score function)**.** The backward drift in (3) contains the following term

$$\nabla\log p_t(x),$$

which is called the Stein's score function in the statistics literature. Once the forward coefficients f and g are fixed, this is the only unknown quantity. So diffusion modeling becomes a score estimation problem.

In practice we do not start the reverse sampler from the exact law p_T. Instead we choose an easy-to-sample approximation p_{noise} and then replace the true score by a learned approximation s_θ. The practical reverse SDE is therefore

$$\mathrm{d}Y_t = \Big(-f(T-t,Y_t) + g(T-t)^2 s_\theta(T-t,Y_t)\Big)\,\mathrm{d}t + g(T-t)\,\mathrm{d}\overline{W}_t, \qquad Y_0 \sim p_{\text{noise}}. \tag{6}$$

The rest of the theory explains how to learn s_θ and how the final sampling error depends on this approximation.

3 Examples: two standard forward diffusions

The choice of forward SDE matters because it determines two things:

- whether p_T is close to a simple noise distribution;
- whether the conditional law of X_t given X_0 is easy enough for score matching.

The most widely used examples are VE and VP.

3.1 Variance exploding (VE) diffusion

Take

$$f(t, x) = 0, \qquad g(t) > 0.$$

Then

$$X_t = X_0 + \int_0^t g(s)\ \mathrm{d}W_s,$$

so conditioned on X_0 we have

$$X_t \mid X_0 \sim \mathcal{N}(X_0, \sigma_t^2 I), \qquad \sigma_t^2 := \int_0^t g(s)^2\ \mathrm{d}s. \tag{7}$$

Thus the forward process simply adds Gaussian noise with growing variance. If σ_T^2 is large, then p_T is close to a broad Gaussian, so we may take $p_{\text{noise}} \approx \mathcal{N}(0, \sigma_T^2 I)$.

The reverse SDE is

$$\mathrm{d}Y_t = g(T-t)^2 \nabla \log p_{T-t}(Y_t)\ \mathrm{d}t + g(T-t)\ \mathrm{d}\overline{W}_t.$$

3.2 Variance preserving (VP) diffusion

Take

$$f(t, x) = -\frac{1}{2}\beta(t)x, \qquad g(t) = \sqrt{\beta(t)}, \qquad \beta(t) > 0.$$

Then the forward SDE is

$$\mathrm{d}X_t = -\frac{1}{2}\beta(t)X_t\ \mathrm{d}t + \sqrt{\beta(t)}\ \mathrm{d}W_t. \tag{8}$$

Define

$$\alpha_t := \exp\Big(-\frac{1}{2}\int_0^t \beta(s)\ \mathrm{d}s\Big).$$

Multiplying (8) by α_t^{-1} and integrating gives

$$X_t = \alpha_t X_0 + \alpha_t \int_0^t \alpha_s^{-1}\sqrt{\beta(s)}\ \mathrm{d}W_s.$$

Hence

$$X_t \mid X_0 \sim \mathcal{N}(\alpha_t X_0, (1-\alpha_t^2)I). \tag{9}$$

So as t grows, the signal is damped by the factor α_t while the variance approaches 1. If α_T is small, then p_T is close to $\mathcal{N}(0, I)$, and we may choose

$$p_{\text{noise}} = \mathcal{N}(0, I).$$

The reverse SDE becomes

$$\mathrm{d}Y_t = \Big(\frac{1}{2}\beta(T-t)Y_t + \beta(T-t)\nabla \log p_{T-t}(Y_t)\Big)\,\mathrm{d}t + \sqrt{\beta(T-t)}\ \mathrm{d}\overline{W}_t.$$

Remark 3.1 (Why VE and VP are so convenient)**.** For both models, $X_t \mid X_0$ is Gaussian. This is exactly what makes denoising score matching simple: the target score of the conditional law is explicit.

4 Learning the score function

The reverse SDE is usable only if we can approximate the score field $\nabla \log p_t$. Let $\{s_\theta(t,x)\}_\theta$ be a family of functions, usually a neural network, intended to approximate this score.

4.1 Implicit score matching

The ideal objective at a fixed time t is

$$\mathbb{E}_{X\sim p_t}\big[\, \|s_\theta(t,X) - \nabla \log p_t(X)\|^2 \,\big].$$

But this involves the unknown score itself. Hyvärinen's key observation is that the loss can be rewritten without the target.

Theorem 4.1 (Implicit score matching). *Fix t, and assume $s_\theta(t,\cdot)$ is smooth and decays fast enough at infinity. Then*

$$\mathbb{E}_{X\sim p_t}\big[\, \|s_\theta(t,X) - \nabla \log p_t(X)\|^2 \,\big] = \mathbb{E}_{X\sim p_t}\big[\, \|s_\theta(t,X)\|^2 + 2\nabla\cdot s_\theta(t,X)\big] + C_t,$$

where C_t is independent of θ. Consequently, minimizing the explicit score-matching loss is equivalent to minimizing

$$\mathbb{E}_{X\sim p_t}\big[\, \|s_\theta(t,X)\|^2 + 2\nabla\cdot s_\theta(t,X)\big].$$

Proof. Expand the square:

$$\mathbb{E}\big[\, \|s_\theta(t,X) - \nabla \log p_t(X)\|^2 \,\big] = \mathbb{E}\big[\, \|s_\theta(t,X)\|^2 \,\big] - 2\mathbb{E}\big[\, \langle s_\theta(t,X), \nabla \log p_t(X)\rangle \,\big] + \mathbb{E}\big[\, \|\nabla \log p_t(X)\|^2 \,\big].$$

The last term does not depend on θ. For the middle term,

$$\begin{aligned}
\mathbb{E}\big[\, \langle s_\theta(t,X), \nabla \log p_t(X)\rangle \,\big] &= \int_{\mathbb{R}^d} s_\theta(t,x)\cdot \nabla p_t(x)\ \mathrm{d}x \\
&= -\int_{\mathbb{R}^d} (\nabla\cdot s_\theta(t,x)) p_t(x)\ \mathrm{d}x \\
&= -\mathbb{E}\big[\nabla\cdot s_\theta(t,X)\big],
\end{aligned}$$

where we used integration by parts. Substituting back gives the claim. □

Implicit score matching removes the unknown target, but it introduces the divergence $\nabla\cdot s_\theta$. In high dimension this is expensive, which is why the denoising formulation is preferred in practice.

4.2 Denoising score matching

We now use the conditional law of X_t given X_0. The key identity is that the score of the marginal is the conditional expectation of the score of the corrupted observation.

Theorem 4.2 (Denoising score matching). *Fix t, and assume the same regularity as above. Then*

$$\mathbb{E}\big[\, \|s_\theta(t,X_t) - \nabla \log p_t(X_t)\|^2 \,\big] = \mathbb{E}\big[\, \|s_\theta(t,X_t) - \nabla_x \log p_t(X_t \mid X_0)\|^2 \,\big] + C'_t,$$

where C'_t is independent of θ. Therefore the explicit score-matching problem is equivalent to minimizing

$$\mathbb{E}\big[\, \|s_\theta(t,X_t) - \nabla_x \log p_t(X_t \mid X_0)\|^2 \,\big].$$

Proof. The crucial identity is Fisher's identity:

$$\nabla \log p_t(x) = \mathbb{E}\big[\nabla_x \log p_t(x \mid X_0) \mid X_t = x\big]. \tag{10}$$

Indeed,

$$p_t(x) = \int p_0(x_0) p_t(x \mid x_0)\, \mathrm{d}x_0,$$

so

$$\nabla p_t(x) = \int p_0(x_0) \nabla_x p_t(x \mid x_0)\, \mathrm{d}x_0.$$

Dividing by $p_t(x)$ gives (10).

Now expand the two quadratic losses. Their $\mathbb{E}\,\|s_\theta(t, X_t)\|^2$ terms are the same, and their final terms differ only by a constant independent of θ. So it is enough to compare the cross terms. Using the tower property and (10),

$$\begin{aligned}
\mathbb{E}\big[\,\langle s_\theta(t, X_t), \nabla \log p_t(X_t)\rangle\,\big] &= \mathbb{E}\Big[\,\langle s_\theta(t, X_t), \mathbb{E}[\nabla_x \log p_t(X_t \mid X_0) \mid X_t]\rangle\,\Big] \\
&= \mathbb{E}\big[\,\langle s_\theta(t, X_t), \nabla_x \log p_t(X_t \mid X_0)\rangle\,\big].
\end{aligned}$$

Hence the two losses differ by a θ-independent constant. $\square$

For the standard models, the conditional law is Gaussian:

$$X_t \mid X_0 \sim \mathcal{N}\big(\mu_t(X_0), \sigma_t^2 I\big).$$

Then

$$\nabla_x \log p_t(x \mid X_0) = -\frac{x - \mu_t(X_0)}{\sigma_t^2}. \tag{11}$$

Therefore the DSM loss becomes

$$\mathbb{E}\left[\left\| s_\theta(t, X_t) + \frac{X_t - \mu_t(X_0)}{\sigma_t^2} \right\|^2\right]. \tag{12}$$

Equivalently,

$$\mathbb{E}\left[\left\| s_\theta(t, X_t) + \frac{X_t - \mu_t(X_0)}{\sigma_t^2} \right\|^2\right] = \frac{1}{\sigma_t^4}\,\mathbb{E}\Big[\,\Big\| X_t + \sigma_t^2 s_\theta(t, X_t) - \mu_t(X_0)\Big\|^2\,\Big].$$

So if we define the associated denoiser by

$$D_\theta(t, x) := x + \sigma_t^2 s_\theta(t, x),$$

then denoising score matching is exactly least-squares regression of $\mu_t(X_0)$ on X_t.

If we write

$$X_t = \mu_t(X_0) + \sigma_t \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I),$$

then (12) is the same as

$$\mathbb{E}\Big[\,\|\sigma_t s_\theta(t, \mu_t(X_0) + \sigma_t \varepsilon) + \varepsilon\|^2\,\Big]. \tag{13}$$

This is the practical training loss used in most score-based diffusion models.

Remark 4.3 (Interpretation)**.** The conditional score in (11) points from the noisy sample X_t back toward its clean mean $\mu_t(X_0)$. Tweedie's formula says the same thing in posterior-mean form:

$$\text{posterior denoiser} = \text{noisy input} + \sigma_t^2 \times \text{score}.$$

So denoising and score estimation are not just related; they are two equivalent ways to package the same information.

4.3 Tweedie's formula

This Gaussian structure also yields a useful posterior-mean identity, usually called Tweedie's formula.

Proposition 4.4 (Tweedie's formula for Gaussian perturbations)**.** *Assume*

$$X_t \mid X_0 \sim \mathcal{N}(\mu_t(X_0), \sigma_t^2 I).$$

Then for every x with $p_t(x) > 0$,

$$\mathbb{E}[\mu_t(X_0) \mid X_t = x] = x + \sigma_t^2 \nabla \log p_t(x). \tag{14}$$

Equivalently,

$$\nabla \log p_t(x) = \frac{\mathbb{E}[\mu_t(X_0) \mid X_t = x] - x}{\sigma_t^2}. \tag{15}$$

Proof. Write the marginal density as

$$p_t(x) = \int_{\mathbb{R}^d} p_0(x_0)\, \varphi_{\sigma_t}(x - \mu_t(x_0))\, \mathrm{d}x_0,$$

where φ_{σ_t} is the density of $\mathcal{N}(0, \sigma_t^2 I)$. Differentiate under the integral sign:

$$\begin{aligned}\nabla p_t(x) &= \int_{\mathbb{R}^d} p_0(x_0)\, \nabla_x \varphi_{\sigma_t}(x - \mu_t(x_0))\, \mathrm{d}x_0 \\ &= -\frac{1}{\sigma_t^2} \int_{\mathbb{R}^d} (x - \mu_t(x_0)) p_0(x_0) \varphi_{\sigma_t}(x - \mu_t(x_0))\, \mathrm{d}x_0.\end{aligned}$$

Dividing by $p_t(x)$ gives

$$\nabla \log p_t(x) = \frac{1}{\sigma_t^2} \frac{\int_{\mathbb{R}^d} (\mu_t(x_0) - x) p_0(x_0) \varphi_{\sigma_t}(x - \mu_t(x_0))\, \mathrm{d}x_0}{\int_{\mathbb{R}^d} p_0(x_0) \varphi_{\sigma_t}(x - \mu_t(x_0))\, \mathrm{d}x_0}.$$

The fraction is exactly $\mathbb{E}[\mu_t(X_0) - x \mid X_t = x]$, which proves (14). □

A few examples:

- For VE, $\mu_t(X_0) = X_0$, so Tweedie's formula becomes

$$\mathbb{E}[X_0 \mid X_t = x] = x + \sigma_t^2 \nabla \log p_t(x).$$

- For VP, $\mu_t(X_0) = \alpha_t X_0$ and $\sigma_t^2 = 1 - \alpha_t^2$, so

$$\mathbb{E}[X_0 \mid X_t = x] = \alpha_t^{-1}\Big(x + (1 - \alpha_t^2) \nabla \log p_t(x)\Big).$$

5 The probability flow ODE

Once the score is known, the reverse SDE is not the only available sampler. There is also a deterministic flow with exactly the same one-time marginals as the forward SDE.

Theorem 5.1 (Probability flow ODE)**.** *Let X_t solve the forward SDE (1), and let p_t be its density. Consider the ODE*

$$\frac{\mathrm{d}\widetilde{X}_t}{\mathrm{d}t} = f(t, \widetilde{X}_t) - \frac{1}{2} g(t)^2 \nabla \log p_t(\widetilde{X}_t), \qquad \widetilde{X}_0 \sim p_{\mathrm{data}}. \tag{16}$$

Then for every $t \in [0, T]$, the random variables X_t and $\widetilde{X}_t$ have the same law.

Proof. Let q_t denote the density of $\widetilde{X}_t$. Since (16) is deterministic, q_t satisfies the continuity equation

$$\partial_t q_t(x) = -\nabla \cdot \Big(\big[f(t,x) - \tfrac{1}{2} g(t)^2 \nabla \log p_t(x) \big] q_t(x) \Big).$$

Now evaluate the right-hand side at $q_t = p_t$:

$$\begin{aligned} -\nabla \cdot \Big(\big[f(t,x) - \tfrac{1}{2} g(t)^2 \nabla \log p_t(x) \big] p_t(x) \Big) &= -\nabla \cdot \big(f(t,x) p_t(x) \big) + \frac{1}{2} g(t)^2 \nabla \cdot \big(p_t(x) \nabla \log p_t(x) \big) \\ &= -\nabla \cdot \big(f(t,x) p_t(x) \big) + \frac{1}{2} g(t)^2 \Delta p_t(x). \end{aligned}$$

This is exactly the Fokker–Planck equation (2). Since $q_0 = p_0$, uniqueness yields $q_t = p_t$ for all t. ☐

Running this dynamics backward gives the reverse-time ODE

$$\frac{\mathrm{d}Y_t}{\mathrm{d}t} = -f(T-t, Y_t) + \frac{1}{2} g(T-t)^2 \nabla \log p_{T-t}(Y_t), \qquad Y_0 \sim p_T. \tag{17}$$

Replacing the true score by s_θ and p_T by p_{noise} yields the practical ODE sampler

$$\frac{\mathrm{d}Y_t}{\mathrm{d}t} = -f(T-t, Y_t) + \frac{1}{2} g(T-t)^2 s_\theta(T-t, Y_t), \qquad Y_0 \sim p_{\mathrm{noise}}. \tag{18}$$

Remark 5.2. The reverse SDE and the probability flow ODE use the same score. The SDE adds Brownian motion; the ODE turns the same marginal evolution into a deterministic flow. This is the continuous-time viewpoint behind DDIM-type samplers.

6 Stability of the reverse sampler

We now study how the final sample quality depends on two errors:

- the initialization error from replacing p_T by p_{noise};
- the score error from replacing $\nabla \log p_t$ by s_θ.

6.1 Total variation stability bound

We begin by controlling the sampling error in total variation distance. This bound is convenient to present first because it follows from a clean decomposition argument together with Girsanov's theorem and Pinsker's inequality. It separates the error into two contributions: the mismatch between the terminal law p_T and the chosen prior p_{noise}, and the score approximation error.

Recall that for two probability measures P and Q on $\mathbb{R}^d$, the total variation distance is defined as

$$d_{\mathrm{TV}}(P,Q) := \sup_{A\in\mathcal{B}(\mathbb{R}^d)} |P(A)-Q(A)|.$$

We consider again the practical reverse SDE

$$\mathrm{d}V_t = \Big(-f(T-t,V_t) + g(T-t)^2 s_\theta(T-t,V_t)\Big)\,\mathrm{d}t + g(T-t)\,\mathrm{d}B_t, \qquad V_0 \sim p_{\mathrm{noise}}. \tag{19}$$

Let U_t denote the exact reverse process

$$\mathrm{d}U_t = \Big(-f(T-t,U_t) + g(T-t)^2\nabla\log p_{T-t}(U_t)\Big)\,\mathrm{d}t + g(T-t)\,\mathrm{d}B_t, \qquad U_0 \sim p_T, \tag{20}$$

so that $\mathrm{Law}(U_T) = p_{\mathrm{data}}$ by Theorem 2.1.

We denote the path-space laws by

$$\mathbb{P}^U := \mathrm{Law}\big((U_t)_{0\le t\le T}\big).$$

Theorem 6.1 (Total variation stability bound)**.** *Assume that for every $t\in[0,T]$,*

$$\mathbb{E}_{X\sim p_t}\big[\,\|s_\theta(t,X)-\nabla\log p_t(X)\|^2\,\big] \le \varepsilon^2.$$

For simplicity assume $g(t)\le 1$. Then

$$d_{\mathrm{TV}}(\mathrm{Law}(V_T),p_{\mathrm{data}}) \le d_{\mathrm{TV}}(p_T,p_{\mathrm{noise}}) + \varepsilon\sqrt{\frac{T}{2}}. \tag{21}$$

Proof. Introduce an intermediate process $\widetilde{V}_t$ defined as (19) but initialized at p_T instead of p_{noise}. By the data processing inequality for total variation,

$$d_{\mathrm{TV}}(\mathrm{Law}(V_T),\mathrm{Law}(U_T)) \le d_{\mathrm{TV}}(p_T,p_{\mathrm{noise}}) + d_{\mathrm{TV}}(\mathrm{Law}(\widetilde{V}_T),\mathrm{Law}(U_T)).$$

It remains to control the second term, which captures the score approximation error. Using Girsanov's theorem, the path measures of $\widetilde{V}$ and U satisfy

$$\mathrm{KL}(\mathbb{P}^U,\mathbb{P}^{\widetilde{V}}) = \frac{1}{2}\,\mathbb{E}\Big[\int_0^T g(T-t)^2\,\|s_\theta(T-t,U_t)-\nabla\log p_{T-t}(U_t)\|^2\,\mathrm{d}t\Big].$$

Since $U_t\sim p_{T-t}$, the assumption implies

$$\mathrm{KL}(\mathbb{P}^U,\mathbb{P}^{\widetilde{V}}) \le \frac{\varepsilon^2}{2}\int_0^T g(T-t)^2\,\mathrm{d}t = \frac{\varepsilon^2}{2}\int_0^T g(t)^2\,\mathrm{d}t.$$

Applying Pinsker's inequality,

$$d_{\mathrm{TV}}(\mathrm{Law}(\widetilde{V}_T),\mathrm{Law}(U_T)) \le \sqrt{\frac{1}{2}\mathrm{KL}(\mathbb{P}^U,\mathbb{P}^{\widetilde{V}})} \le \frac{\varepsilon}{2}\Big(\int_0^T g(t)^2\,\mathrm{d}t\Big)^{1/2}.$$

Combining the above estimates yields (21).

□

Remark 6.2. (a) The bound separates two sources of error:

$$\text{total error} \lesssim \text{initialization mismatch} + \text{score approximation error}.$$

(b) The first term depends only on how well p_{noise} approximates p_T, while the second grows at most like $\sqrt{T}$.

(c) The total variation estimate is dimension-free and relies only on a path-space change-of-measure argument (Girsanov + Pinsker). As a result, it does not require any regularity (e.g., Lipschitz continuity) of the learned score.

(d) The price to pay is that total variation is a strong metric: it detects any discrepancy in support and does not exploit the geometry of the state space. Consequently, the bound is primarily qualitative—it guarantees correctness of the sampling procedure, but does not directly quantify how far typical samples are from the target.

Corollary 6.3 (Total variation bound for VP diffusion)**.** *Assume the forward process is the variance preserving (VP) SDE*

$$\mathrm{d}X_t = -\tfrac{1}{2}\beta(t)X_t \,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}W_t,$$

and let V_t be the corresponding approximate reverse process (19). Assume:

(i) the score error satisfies

$$\mathbb{E}_{X\sim p_t}\big[\,\|s_\theta(t,X) - \nabla \log p_t(X)\|^2\,\big] \le \varepsilon^2;$$

(ii) $\beta(t)$ is bounded away from zero;

(iii) $\mathbb{E}_{X_0\sim p_{\text{data}}}\|X_0\|^2 < \infty$.

(iv) $p_{noise} = \mathcal{N}(0,I)$

Then for T sufficiently large,

$$d_{\text{TV}}(\text{Law}(V_T), p_{\text{data}}) \le e^{-\frac{1}{2}\int_0^T \beta(s)\,\mathrm{d}s}\sqrt{\frac{1}{2}\mathbb{E}_{X_0\sim p_{\text{data}}}\|X_0\|^2} + \varepsilon\sqrt{\frac{T}{2}}. \tag{22}$$

Proof. By Theorem 6.1, we already have

$$d_{\text{TV}}(\text{Law}(V_T), p_{\text{data}}) \le d_{\text{TV}}(p_T, p_{\text{noise}}) + \varepsilon\sqrt{\tfrac{T}{2}}.$$

It remains to bound the initialization error $d_{\text{TV}}(p_T, p_{\text{noise}})$.

For the VP diffusion, the conditional law is

$$X_T \mid X_0 \sim \mathcal{N}\big(\alpha_T X_0, (1-\alpha_T^2)I\big), \quad \alpha_T := \exp\Big(-\tfrac{1}{2}\int_0^T \beta(s)\,\mathrm{d}s\Big).$$

We take $p_{\text{noise}} = \mathcal{N}(0,I)$.

Using convexity of KL divergence,

$$\text{KL}(p_T, p_{\text{noise}}) \le \mathbb{E}_{X_0\sim p_{\text{data}}}\Big[\text{KL}\big(\mathcal{N}(\alpha_T X_0, (1-\alpha_T^2)I), \mathcal{N}(0,I)\big)\Big].$$

For Gaussian measures, we compute explicitly

$$\mathrm{KL}(\mathcal{N}(\alpha_T x, (1-\alpha_T^2)I), \mathcal{N}(0,I)) = \frac{1}{2}\Big(\alpha_T^2 \|x\|^2 - d(\alpha_T^2 + \log(1-\alpha_T^2))\Big).$$

As $T \to \infty$, $\alpha_T \to 0$, hence

$$\mathrm{KL}(p_T, p_{\text{noise}}) \le \frac{1}{2}\alpha_T^2 \mathbb{E}\|X_0\|^2.$$

Applying Pinsker's inequality,

$$d_{\mathrm{TV}}(p_T, p_{\text{noise}}) \le \sqrt{\tfrac{1}{2}\mathrm{KL}(p_T, p_{\text{noise}})} \le \alpha_T \sqrt{\frac{1}{2}\mathbb{E}\|X_0\|^2}.$$

Substituting $\alpha_T = \exp\big(-\frac{1}{2}\int_0^T \beta(s)\,\mathrm{d}s\big)$ and combining with the TV decomposition yields (22).

$\square$

6.2 Wasserstein bound

We next establish a stability bound in Wasserstein distance. Unlike total variation, the Wasserstein metric takes into account the geometry of the state space and is therefore better suited for quantitative comparisons when all distributions have finite second moments. The proof uses a synchronous coupling of the exact and approximate reverse processes and yields a differential inequality for the mean-square error.

We assume all distributions below have finite second moments.

Theorem 6.4 (Wasserstein stability bound)**.** *Consider the practical reverse SDE*

$$\mathrm{d}V_t = \Big(-f(T-t, V_t) + g(T-t)^2 s_\theta(T-t, V_t)\Big)\,\mathrm{d}t + g(T-t)\,\mathrm{d}B_t, \qquad V_0 \sim p_{\text{noise}}. \tag{23}$$

Assume:

(i) for every $t \in [0,T]$,

$$\mathbb{E}_{X\sim p_t}\big[\,\|s_\theta(t,X) - \nabla \log p_t(X)\|^2\,\big] \le \varepsilon^2;$$

(ii) there exists $r_f : [0,T] \to \mathbb{R}$ such that

$$\langle x-y, f(t,x) - f(t,y)\rangle \ge r_f(t)\,\|x-y\|^2 \qquad \textit{for all } x,y \in \mathbb{R}^d;$$

(iii) there exists $L > 0$ such that

$$\|s_\theta(t,x) - s_\theta(t,y)\| \le L\,\|x-y\| \qquad \textit{for all } x,y \in \mathbb{R}^d.$$

Fix $h > 0$, and define

$$u(t) := \int_{T-t}^{T} \Big(-2r_f(s) + (2L+2h)g(s)^2\Big)\,\mathrm{d}s. \tag{24}$$

Then

$$W_2\big(\mathrm{Law}(V_T), p_{\text{data}}\big) \le \left[e^{u(T)} W_2(p_T, p_{\text{noise}})^2 + \frac{\varepsilon^2}{2h}\int_0^T g(t)^2 e^{u(T)-u(T-t)}\,\mathrm{d}t\right]^{1/2}. \tag{25}$$

Proof. Let U_t be the exact reverse process started from the correct terminal law:

$$\mathrm{d}U_t = \Big(- f(T-t, U_t) + g(T-t)^2 \nabla \log p_{T-t}(U_t) \Big)\, \mathrm{d}t + g(T-t)\ \mathrm{d}B_t, \qquad U_0 \sim p_T. \tag{26}$$

By Theorem 2.1,

$$\mathrm{Law}(U_t) = p_{T-t} \qquad \text{for every } t,$$

and in particular $U_T \sim p_{\text{data}}$.

Now couple U_0 and V_0 optimally, so that

$$\mathbb{E} \left\| U_0 - V_0 \right\|^2 = W_2(p_T, p_{\text{noise}})^2,$$

and drive both SDEs with the *same* Brownian motion B_t. This is the synchronous coupling. Set

$$D_t := U_t - V_t, \qquad m(t) := \mathbb{E} \left\| D_t \right\|^2.$$

Because the diffusion terms cancel, D_t satisfies the random ODE

$$\mathrm{d}D_t = \Big(- f(T-t, U_t) + f(T-t, V_t) + g(T-t)^2 \left[\nabla \log p_{T-t}(U_t) - s_\theta(T-t, V_t) \right] \Big)\, \mathrm{d}t.$$

Hence

$$\begin{aligned} m'(t) = &-2\mathbb{E} \left\langle D_t, f(T-t, U_t) - f(T-t, V_t) \right\rangle \\ &+ 2g(T-t)^2 \mathbb{E} \left\langle D_t, \nabla \log p_{T-t}(U_t) - s_\theta(T-t, V_t) \right\rangle. \end{aligned}$$

We estimate the two pieces separately.

First, by assumption (ii),

$$-2\mathbb{E} \left\langle D_t, f(T-t, U_t) - f(T-t, V_t) \right\rangle \le -2r_f(T-t) m(t).$$

Second, split the score term as

$$\begin{aligned} &\nabla \log p_{T-t}(U_t) - s_\theta(T-t, V_t) \\ &\quad = \big(\nabla \log p_{T-t}(U_t) - s_\theta(T-t, U_t) \big) + \big(s_\theta(T-t, U_t) - s_\theta(T-t, V_t) \big). \end{aligned}$$

For the first part, Young's inequality $2 \langle a, b \rangle \le 2h \left\| a \right\|^2 + \frac{1}{2h} \left\| b \right\|^2$ gives

$$\begin{aligned} &2g(T-t)^2 \mathbb{E} \left\langle D_t, \nabla \log p_{T-t}(U_t) - s_\theta(T-t, U_t) \right\rangle \\ &\quad \le 2hg(T-t)^2 m(t) + \frac{g(T-t)^2}{2h} \mathbb{E} \left\| \nabla \log p_{T-t}(U_t) - s_\theta(T-t, U_t) \right\|^2. \end{aligned}$$

But $U_t \sim p_{T-t}$, so assumption (i) implies

$$\mathbb{E} \left\| \nabla \log p_{T-t}(U_t) - s_\theta(T-t, U_t) \right\|^2 \le \varepsilon^2.$$

Therefore

$$2g(T-t)^2 \mathbb{E} \left\langle D_t, \nabla \log p_{T-t}(U_t) - s_\theta(T-t, U_t) \right\rangle \le 2hg(T-t)^2 m(t) + \frac{\varepsilon^2}{2h} g(T-t)^2. \tag{27}$$

For the second part, the Lipschitz property (iii) yields

$$\begin{aligned}2g(T-t)^2\mathbb{E}\,\langle D_t, s_\theta(T-t,U_t)-s_\theta(T-t,V_t)\rangle &\le 2g(T-t)^2\mathbb{E}\Big[\,\|D_t\|\,\|s_\theta(T-t,U_t)-s_\theta(T-t,V_t)\|\,\Big]\\ &\le 2Lg(T-t)^2 m(t).\end{aligned}$$

Combining the above bounds, we get the differential inequality

$$m'(t) \le \Big(-2r_f(T-t)+(2L+2h)g(T-t)^2\Big)m(t)+\frac{\varepsilon^2}{2h}g(T-t)^2.$$

Define

$$a(t) := -2r_f(T-t)+(2L+2h)g(T-t)^2.$$

Then Grönwall's inequality gives

$$m(T) \le e^{\int_0^T a(s)\,\mathrm{d}s}m(0)+\frac{\varepsilon^2}{2h}\int_0^T g(T-s)^2 e^{\int_s^T a(r)\,\mathrm{d}r}\,\mathrm{d}s.$$

By definition of u,

$$\int_0^t a(s)\,\mathrm{d}s = u(t).$$

Hence

$$m(T) \le e^{u(T)}W_2(p_T,p_{\mathrm{noise}})^2+\frac{\varepsilon^2}{2h}\int_0^T g(T-s)^2 e^{u(T)-u(s)}\,\mathrm{d}s.$$

Changing variables $t=T-s$ in the integral gives

$$m(T) \le e^{u(T)}W_2(p_T,p_{\mathrm{noise}})^2+\frac{\varepsilon^2}{2h}\int_0^T g(t)^2 e^{u(T)-u(T-t)}\,\mathrm{d}t.$$

Finally, since $U_T \sim p_{\mathrm{data}}$,

$$W_2\big(\mathrm{Law}(V_T),p_{\mathrm{data}}\big)^2 \le \mathbb{E}\,\|U_T-V_T\|^2 = m(T).$$

Taking square roots yields (25). □

Remark 6.5. In contrast to total variation, the Wasserstein estimate reflects the geometry of the reverse dynamics. The error propagation depends explicitly on:

- the contractivity of the drift through r_f, and the Lipschitz constant of the learned score.

When the reverse dynamics are contractive (e.g., strongly dissipative drift), the exponential factor $e^{u(T)}$ remains controlled, and the score error accumulates in a stable manner. Otherwise, the bound may deteriorate, reflecting genuine instability of the reverse flow.

Remark 6.6 (Total variation vs. Wasserstein bounds)**.** The two stability estimates capture different aspects of the sampling error:

Total variation bound (Theorem 6.1):

- Requires minimal assumptions (no Lipschitz or contractivity conditions);
- Dimension-free and robust;

- Provides a worst-case guarantee, but is insensitive to the geometry of the state space.

Wasserstein bound (Theorem 6.4):

- Requires additional structure (monotonicity of Lipschitz score);
- Captures metric error and is therefore more informative for sample quality;
- Reveals how errors propagate along the reverse dynamics through a Grönwall-type estimate.

In particular, when the reverse dynamics are contractive, the Wasserstein bound can yield sharper and more quantitative guarantees, while the total variation bound remains valid in much greater generality. The two results should therefore be viewed as complementary: TV provides robustness, whereas Wasserstein provides quantitative control.

7 Takeaways

The logic of score-based diffusion models can now be summarized in one chain:

1. Choose a forward diffusion that gradually transforms data into something close to Gaussian noise.
2. Use time reversal to identify the backward drift. The new term is the score $\nabla \log p_t$.
3. Learn that score by denoising score matching, which is practical because $X_t \mid X_0$ is Gaussian for the standard models.
4. Sample either from the reverse SDE or from the probability flow ODE.
5. Control the final error by separately bounding the initialization mismatch $p_T \leftrightarrow p_{\text{noise}}$ and the score approximation error.

This is the basic analytic backbone of diffusion modeling in continuous time.

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